

Exam 3 Guide

See the Topics Schedule and Assessment sections in the Syllabus for exam schedules and procedures.

Topics

1. Basics

- **1st-order System of ODEs forms.** Linear vs non-linear, and homogeneous vs non-homogeneous
- **Matrix-vector form.** Converting a system of ODEs to matrix-vector form
- **Types of solutions.** General, specific, homogeneous, and particular
- **Verifying solutions.** 1st-order system of ODEs

2. Modelling Physical Systems

- **Interpreting 1st-order system of ODEs.** Making meaningful 1st-order linear systems of ODEs
- **Spring-mass system.** Damped and undamped systems
- **Two-tank system.** Homogenous and non-homogeneous systems
- **Population dynamics.** Linear and Non-linear systems

3. Qualitative Analysis of 1st-Order Linear System of ODEs

- **Existence and uniqueness.** Initial value problems and equilibriums
- **Linearization.** Using the Jacobian matrix to predict solution behaviors near an equilibrium
- **Equilibrium solutions and stability.** Node, saddle, spiral, center, and fixed points
- **Phase planes.** An $x-y$ plane representing two dependent variables of a system of ODEs
- **Vector fields.** Slope vectors representing the rate of change on a phase plane
- **Phase portraits.** A vector field with specific solution trajectories

4. Solving 1st-Order Linear System of ODEs

- **Homogeneous solutions.** Characteristic equations and eigensolutions in vector form:
 - **Eigenvalues.** Finding eigenvalues and use them to identify the stability of equilibriums
 - **Eigenvectors.** Finding eigenvectors and use them to determine the eigensolutions
- **Non-Homogeneous solutions.** Particular solutions as vector functions using the following methods:
 - **Undetermined coefficients.** Create a trial solution and solve the coefficients in vector form
 - **Variation of parameters.** Using the homogeneous solutions and the Wronskian in vector form
- **Graphical.** Solving for the isoclines and nulleclines